CSE225 DATA STRUCTURES **Student Name:_____**
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Q1. (25pts)
Consider the following piece of code!

```
...
long res=0;
...
for (i=1; i<=n; i++)
  for (j=1; j<=i; j*=2)
    res += (i+j);
printf("res=%d", res);
...
```

Figure 1A: Code for part (a)

a) (10pts)
What is the value of res, **in terms of *n***, printed out at the printf statement in Fig. 1A? **Show all your work!**

$$S=\sum_{i=1}^{n}\sum_{j=1}^{\lfloor \log_2 i \rfloor+1} i+2^{j-1}=\sum_{i=1}^{n}(\lfloor \log_2 i \rfloor+1)i+2^{\lfloor \log_2 i \rfloor+1}-1$$

n	1	2	3	4	5	6	7	8	…	k	
res	2	9	18	37	59	84	112	159	…	$\sum_{i=1}^{k}\sum_{j=1}^{\lfloor \log_2 i \rfloor+1}\left[i+2^{j-1}\right]$	

n **res**
1 2
2 2+2*2+(1+2)
3 9+2*3+(1+2)
4 18+3*4+(1+2+4)
5,…,7 res(i-1)+2*i+(1+2+4) where $i \in \{5,6,7\}$
8 res(7)+3*8+(1+2+4+8)
…
k $res(k-1)+k*(\lfloor \log_2 k \rfloor+1)+2^{\lfloor \log_2 k \rfloor+1}-1;$

$$res=\sum_{i=1}^{n}\sum_{j=1}^{\lfloor \log_2 i \rfloor+1}\left[i+2^{j-1}\right];$$